DEBT SUSTAINABILITY ANALYSIS (DSA)—UPDATE

The DSA update indicates that Moldova's risk of debt distress remains low and overall public sector debt dynamics are sustainable.[1] *That said, vulnerabilities have risen since the previous assessment.*

1. **The worsening external environment led to a downward shift in key macroeconomic projections in the near term compared to the previous assessment.**[2] The slowdown in global growth, in particular, in the European Union, is expected to affect Moldova's economy via trade, remittances, and private capital flow channels, reducing projected GDP growth. However, the relatively resilient domestic consumption is likely to result in more gradual adjustment in the current account deficit over the medium term than previously expected.

Table. Moldova: Evolution of selected macroeconomic indicators

	2010	2011	2012	2013	2014	2015
Real GDP growth (%)						
Previous DSA 1/	6.9	5.0	5.0	5.0	4.5	4.5
Current DSA	7.1	6.4	3.0	5.0	5.0	5.5
Nominal GDP (US$ million)						
Previous DSA 1/	5,810	6,999	7,680	8,433	9,217	10,072
Current DSA	5,813	7,003	7,589	8,216	8,969	9,776
Primary fiscal deficit (% of GDP)						
Previous DSA 1/	1.7	1.1	0.0	-0.1	-0.1	-0.1
Current DSA	1.7	1.6	0.1	0.1	0.0	0.0
Non-interest current account deficit (% of GDP)						
Previous DSA 1/	7.4	7.7	8.4	8.0	6.7	6.0
Current DSA	7.0	10.7	9.8	9.1	8.1	7.7

1/ Please see IMF Country Report No. 11/200.

2. **The debt dynamics are largely in line with previous projections.** The public and publicly guaranteed (PPG) debt stood at 27.8 percent of GDP as of end-2011 compared to 29.4 percent at end-2010. The decline in the PPG debt-to-GDP ratio came on the back of brisk GDP growth, relatively stable exchange rate, and restrained borrowing by the central government. Most of the PPG debt was external and to multilateral creditors. The domestic PPG debt came at 6.8 percent of GDP at end-2011, declining from 7.4 percent a year ago, as the budget deficit was financed primarily

[1] The DSA update is conducted jointly by World Bank and Fund staffs in line with the Guidance Note on the Application of the Joint Bank-Fund Debt Sustainability Framework for Low-Income Countries (January 2010).

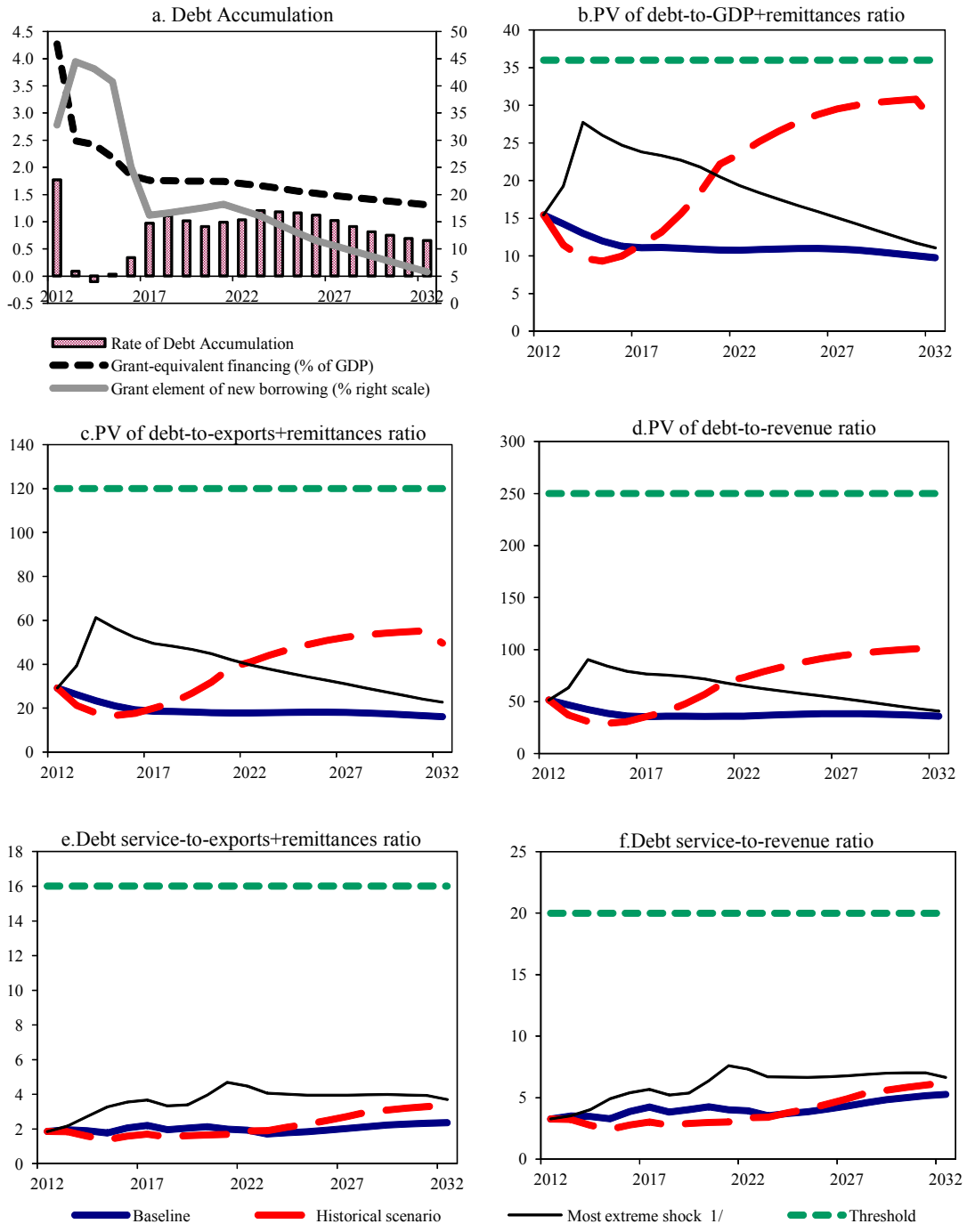
[2] The previous DSA was conducted in July, 2011 (IMF Country Report No. 11/200).

from external sources. Over the medium- to long-term, the projections for government borrowing are broadly unchanged.

3. **While the DSA framework indicates a low risk of public external debt distress, other indicators suggest that vulnerabilities have risen lately**. All external debt indicators remain well below the thresholds under the standard bound tests and alternative scenarios. The public DSA suggests that the overall public sector debt dynamics are sustainable unless there is a substantial and prolonged adverse growth shock. Nevertheless, the authorities—both central and local governments—should continue to be prudent with their borrowing strategy, given Moldova's limited debt-servicing capacity. In addition, a shift in the large private sector external debt (45 percent of GDP) to shorter maturities, continuing rise in the already large stock of gas payment arrears by the breakaway region of Transnistria,[3] and substantial fiscal risks related to possible claims on the budget for compensation for judicial errors/inaction pose risks to the debt position.

[3] For an analysis of the impact of Transnistria's debt on Moldova's debt vulnerabilities, see IMF Country Report No. 11/200.

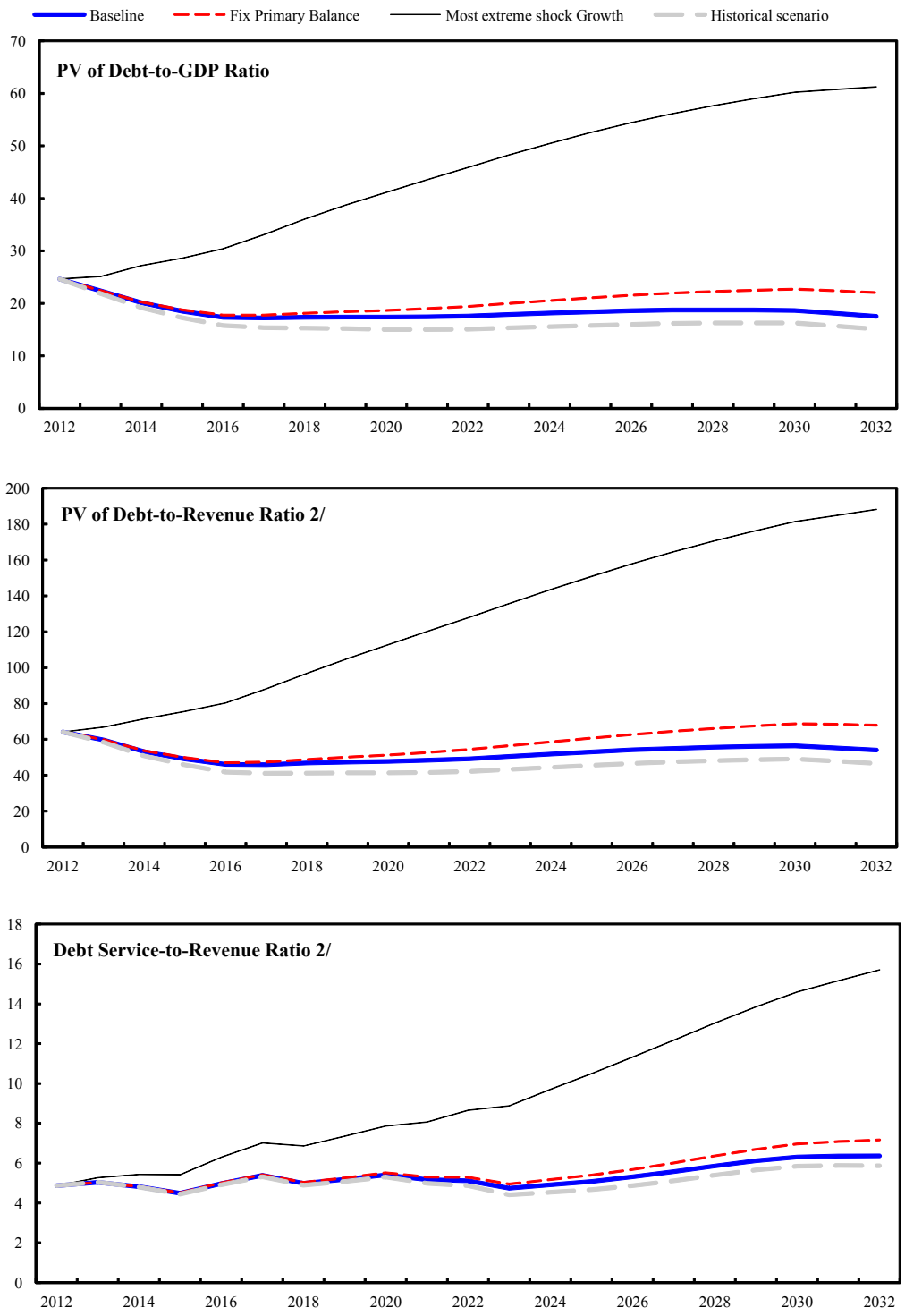
Figure 1. Moldova: Indicators of Public and Publicly Guaranteed External Debt under Alternatives Scenarios, 2012-2032 1/



Sources: Country authorities; and staff estimates and projections.

1/ The most extreme stress test is the test that yields the highest ratio in 2022. In figure b. it corresponds to a Exports shock; in c. to a Exports shock; in d. to a Exports shock; in e. to a Exports shock and in figure f. to a Exports shock

Figure 2.Moldova: Indicators of Public Debt Under Alternative Scenarios, 2012-2032 1/



Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in 2022.
2/ Revenues are defined inclusive of grants.

Table 1a.Moldova: Public Sector Debt Sustainability Framework, Baseline Scenario, 2009-2032
(In percent of GDP, unless otherwise indicated)

	Actual					Estimate						Projections			
	2009	2010	2011	Average 5/	Standard Deviation 5/	2012	2013	2014	2015	2016	2017	2012-17 Average	2022	2032	2018-32 Average
Public sector debt 1/	31.2	29.4	27.8			28.3	25.9	23.3	21.6	20.3	20.0		20.7	20.1	
o/w foreign-currency denominated	22.7	22.1	21.0			21.8	20.1	18.4	17.0	16.0	15.6		15.5	13.8	
Change in public sector debt	10.0	-1.7	-1.7			0.6	-2.5	-2.5	-1.7	-1.3	-0.2		0.2	-0.6	
Identified debt-creating flows	9.9	-3.2	-2.4			-1.5	-2.0	-1.6	-1.4	-1.2	-1.2		-1.2	-1.2	
Primary deficit	4.9	1.7	1.6	-0.5	2.8	0.1	0.1	0.0	0.0	0.0	0.0	0.0	-0.2	-0.3	-0.2
Revenue and grants	38.9	38.3	36.7			38.4	37.5	37.8	37.7	37.7	37.4		35.7	32.4	
of which: grants	2.1	2.8	2.1			3.2	2.0	2.0	1.7	1.5	1.4		1.3	1.2	
Primary (noninterest) expenditure	43.8	40.0	38.3			38.6	37.6	37.8	37.7	37.7	37.4		35.4	32.1	
Automatic debt dynamics	5.3	-4.4	-3.5			-1.2	-1.7	-1.5	-1.4	-1.2	-1.2		-0.9	-1.0	
Contribution from interest rate/growth differential	2.5	-2.2	-1.9			-0.7	-1.2	-1.1	-1.1	-1.0	-0.9		-0.7	-0.8	
of which: contribution from average real interest rate	1.1	-0.2	-0.1			0.1	0.1	0.2	0.1	0.1	0.1		0.3	0.2	
of which: contribution from real GDP growth	1.3	-2.1	-1.8			-0.8	-1.3	-1.2	-1.2	-1.1	-1.0		-1.0	-1.0	
Contribution from real exchange rate depreciation	2.9	-2.2	-1.7			-0.6	-0.5	-0.5	-0.3	-0.3	-0.3		...	...	
Other identified debt-creating flows	-0.4	-0.5	-0.4			-0.4	-0.4	-0.1	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	-0.4	-0.5	-0.4			-0.4	-0.4	-0.1	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	0.2	1.5	0.7			2.1	-0.4	-0.9	-0.4	-0.1	1.0		1.3	0.6	
Other Sustainability Indicators															
PV of public sector debt	...	...	24.5			24.6	22.4	20.1	18.5	17.3	17.2		17.6	17.5	
o/w foreign-currency denominated	...	...	17.7			18.1	16.7	15.2	13.9	13.1	12.8		12.4	11.3	
o/w external	...	...	17.7			18.1	16.7	15.2	13.9	13.1	12.8		12.4	11.3	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	10.2	9.8	9.5			7.9	7.7	6.9	6.0	5.8	5.6		5.7	4.8	
PV of public sector debt-to-revenue and grants ratio (in percent)	…	…	66.7			64.1	59.7	53.2	49.1	46.0	46.0		49.3	54.1	
PV of public sector debt-to-revenue ratio (in percent)	…	…	70.7			70.0	63.0	56.1	51.5	47.8	47.7		51.2	56.2	
o/w external 3/	…	…	51.0			51.4	46.9	42.4	38.8	36.2	35.6		36.1	36.1	
Debt service-to-revenue and grants ratio (in percent) 4/	6.9	4.8	5.5			4.9	5.0	4.8	4.5	5.0	5.4		5.1	6.4	
Debt service-to-revenue ratio (in percent) 4/	7.3	5.2	5.8			5.3	5.3	5.1	4.7	5.2	5.6		5.3	6.6	
Primary deficit that stabilizes the debt-to-GDP ratio	-5.1	3.4	3.2			-0.4	2.6	2.5	1.8	1.3	0.2		-0.4	0.4	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	-6.0	7.1	6.4	4.7	4.6	3.0	5.0	5.0	5.5	5.3	5.3	4.9	5.0	5.0	5.0
Average nominal interest rate on forex debt (in percent)	2.2	1.5	1.2	2.3	0.9	1.2	1.1	1.3	1.3	1.4	1.7	1.3	2.8	3.3	2.8
Average real interest rate on domestic debt (in percent)	15.7	-3.4	1.5	1.3	7.3	2.3	3.1	3.5	3.6	3.7	4.4	3.4	3.3	0.8	2.7
Real exchange rate depreciation (in percent, + indicates depreciation)	17.2	-10.1	-8.2	-7.4	11.4	-2.7	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	2.2	11.1	7.4	9.6	4.1	6.5	5.0	5.0	5.0	5.0	5.0	5.3	5.0	5.0	5.0
Growth of real primary spending (deflated by GDP deflator, in percent)	0.0	0.0	0.0	0.1	0.1	0.0	0.0	0.1	0.1	0.1	0.0	0.0	0.0	0.0	0.0
Grant element of new external borrowing (in percent)	...	...	...	…	…	32.8	44.5	43.2	40.8	25.1	16.2	33.8	17.1	5.8	...

Sources: Country authorities; and staff estimates and projections.
1/ The public debt represents central government direct and guaranteed debt and National Bank of Moldova's borrowing from the IMF on the gross basis.
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
3/ Revenues excluding grants.
4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2a.Moldova: Sensitivity Analysis for Key Indicators of Public Debt 2012-2032

	Projections							
	2012	2013	2014	2015	2016	2017	2022	2032
PV of Debt-to-GDP Ratio								
Baseline	25	22	20	19	17	17	18	18
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	25	22	19	17	16	15	15	15
A2. Primary balance is unchanged from 2012	25	22	20	19	18	18	19	22
A3. Permanently lower GDP growth 1/	25	23	21	21	21	23	36	76
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2013-2014	25	25	27	29	30	33	46	61
B2. Primary balance is at historical average minus one standard deviations in 2013-2014	25	25	24	23	21	21	20	19
B3. Combination of B1-B2 using one half standard deviation shocks	25	24	23	23	23	24	31	39
B4. One-time 30 percent real depreciation in 2013	25	30	27	25	23	23	22	22
B5. 10 percent of GDP increase in other debt-creating flows in 2013	25	32	29	27	26	25	24	21
PV of Debt-to-Revenue Ratio 2/								
Baseline	64	60	53	49	46	46	49	54
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	64	58	51	46	42	41	42	47
A2. Primary balance is unchanged from 2012	64	60	54	50	47	47	54	68
A3. Permanently lower GDP growth 1/	64	61	57	56	56	61	100	233
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2013-2014	64	67	72	76	80	88	128	188
B2. Primary balance is at historical average minus one standard deviations in 2013-2014	64	65	64	60	56	56	57	60
B3. Combination of B1-B2 using one half standard deviation shocks	64	63	60	60	61	65	88	121
B4. One-time 30 percent real depreciation in 2013	64	79	71	66	61	60	61	68
B5. 10 percent of GDP increase in other debt-creating flows in 2013	64	85	77	72	68	67	67	66
Debt Service-to-Revenue Ratio 2/								
Baseline	5	5	5	4	5	5	5	6
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	5	5	5	4	5	5	5	6
A2. Primary balance is unchanged from 2012	5	5	5	5	5	5	5	7
A3. Permanently lower GDP growth 1/	5	5	5	5	5	6	7	17
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2013-2014	5	5	5	5	6	7	9	16
B2. Primary balance is at historical average minus one standard deviations in 2013-2014	5	5	5	5	5	6	6	7
B3. Combination of B1-B2 using one half standard deviation shocks	5	5	5	5	6	6	7	11
B4. One-time 30 percent real depreciation in 2013	5	6	6	6	7	7	7	10
B5. 10 percent of GDP increase in other debt-creating flows in 2013	5	5	6	6	6	6	7	8

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

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Table 3a.: External Debt Sustainability Framework, Baseline Scenario, 2009-2032 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2009	2010	2011			2012	2013	2014	2015	2016	2017	2012-2017 Average	2022	2032	2018-2032 Average
External debt (nominal) 1/	**67.8**	**66.6**	**64.6**			**69.9**	**71.9**	**71.5**	**71.6**	**72.0**	**71.1**		**75.9**	**53.0**	
o/w public and publicly guaranteed (PPG)	22.7	22.1	21.0			21.8	20.1	18.4	17.0	16.0	15.6		15.5	13.8	
Change in external debt	12.6	-1.1	-2.0			5.3	2.0	-0.4	0.1	0.4	-0.8		-2.5	-6.0	
Identified net debt-creating flows	12.1	0.2	-3.8			5.5	3.0	1.3	0.1	-0.6	-1.2		-2.7	-2.5	
Non-interest current account deficit	**7.2**	**7.0**	**10.7**	**8.2**	**5.0**	**9.8**	**9.1**	**8.1**	**7.7**	**7.4**	**7.0**		**5.1**	**5.2**	5.3
Deficit in balance of goods and services	36.6	39.4	41.3			40.9	40.8	39.1	38.5	37.9	37.6		34.1	33.8	
Exports	36.8	39.4	45.0			44.7	46.6	48.3	49.9	51.7	53.1		54.3	54.3	
Imports	73.4	78.8	86.3			85.6	87.4	87.4	88.4	89.6	90.7		88.5	88.1	
Net current transfers (negative = inflow)	-21.2	-21.0	-20.6	-20.9	4.3	-20.1	-19.7	-19.0	-18.5	-18.1	-17.8		-17.3	-17.3	-17.3
o/w official	-1.9	-2.4	-2.1			-2.7	-2.6	-2.3	-2.1	-1.8	-1.8		-1.8	-1.8	
Other current account flows (negative = net inflow)	-8.2	-11.4	-10.0			-11.1	-12.0	-12.0	-12.3	-12.4	-12.8		-11.7	-11.2	
Net FDI (negative = inflow)	**-2.7**	**-3.4**	**-3.9**	**-6.1**	**3.7**	**-4.1**	**-4.4**	**-5.5**	**-5.9**	**-6.3**	**-6.5**		**-6.0**	**-6.3**	-6.1
Endogenous debt dynamics 2/	**7.6**	**-3.4**	**-10.5**			**-0.2**	**-1.7**	**-1.3**	**-1.7**	**-1.7**	**-1.7**		**-1.8**	**-1.4**	
Contribution from nominal interest rate	1.3	1.0	0.8			1.6	1.5	2.0	1.9	1.8	1.7		1.8	1.3	
Contribution from real GDP growth	3.7	-4.5	-3.5			-1.8	-3.2	-3.3	-3.6	-3.5	-3.5		-3.6	-2.7	
Contribution from price and exchange rate changes	2.6	0.1	-7.8			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	**0.4**	**-1.3**	**1.8**			**-0.1**	**-1.1**	**-1.7**	**-0.1**	**0.9**	**0.3**		**0.2**	**-3.5**	
o/w exceptional financing	-2.3	-2.4	-3.0			-3.5	-2.0	-1.5	-1.2	-0.9	-0.9		-0.3	-0.1	
PV of external debt 4/	…	…	61.3			66.3	68.5	68.3	68.6	69.0	68.3		72.8	50.5	
In percent of exports	…	…	136.2			148.2	147.1	141.5	137.4	133.5	128.6		134.0	92.9	
PV of PPG external debt	**…**	**…**	**17.7**			**18.1**	**16.7**	**15.2**	**13.9**	**13.1**	**12.8**		**12.4**	**11.3**	
In percent of exports	**…**	**…**	**39.2**			**40.5**	**35.8**	**31.4**	**27.9**	**25.4**	**24.2**		**22.8**	**20.7**	
In percent of government revenues	**…**	**…**	**51.0**			**51.4**	**46.9**	**42.4**	**38.8**	**36.2**	**35.6**		**36.1**	**36.1**	
Debt service-to-exports ratio (in percent)	**19.9**	**18.6**	**14.9**			**16.0**	**13.0**	**14.9**	**15.3**	**14.5**	**15.3**		**14.1**	**10.4**	
PPG debt service-to-exports ratio (in percent)	**4.4**	**3.3**	**2.8**			**2.6**	**2.7**	**2.6**	**2.4**	**2.7**	**2.9**		**2.5**	**3.0**	
PPG debt service-to-revenue ratio (in percent)	**4.4**	**3.7**	**3.6**			**3.3**	**3.5**	**3.5**	**3.3**	**3.9**	**4.2**		**3.9**	**5.3**	
Total gross financing need (Millions of U.S. dollars)	2000.5	2014.7	2487.1			2829.2	3164.8	3588.5	3918.6	4230.5	4716.8		8443.4	14166.4	
Non-interest current account deficit that stabilizes debt ratio	-5.3	8.1	12.7			4.4	7.1	8.5	7.6	7.1	7.9		7.6	11.2	
Key macroeconomic assumptions															
Real GDP growth (in percent)	-6.0	7.1	6.4	4.7	4.6	3.0	5.0	5.0	5.5	5.3	5.3	4.9	5.0	5.0	5.0
GDP deflator in US dollar terms (change in percent)	-4.5	-0.2	13.2	12.5	11.8	5.2	3.1	4.0	3.3	3.4	4.2	3.9	3.5	3.5	3.5
Effective interest rate (percent) 5/	2.1	1.5	1.4	2.1	0.6	2.7	2.4	3.0	3.0	2.7	2.7	2.7	2.4	2.4	2.5
Growth of exports of G&S (US dollar terms, in percent)	-19.7	14.6	37.5	16.1	18.3	7.7	12.8	13.2	12.7	12.8	12.7	12.0	8.7	8.7	8.9
Growth of imports of G&S (US dollar terms, in percent)	-30.1	14.8	31.9	19.2	21.6	7.5	10.5	9.1	10.2	10.3	11.1	9.8	8.2	8.7	8.5
Grant element of new public sector borrowing (in percent)	…	…	…	…	…	32.8	44.5	43.2	40.8	25.1	16.2	33.8	17.1	5.8	12.7
Government revenues (excluding grants, in percent of GDP)	36.8	35.5	34.6			35.2	35.6	35.8	36.0	36.3	36.0		34.4	31.2	33.4
Aid flows (in Millions of US dollars) 7/	137.3	225.6	200.3			299.9	216.4	222.8	208.7	189.4	197.8		274.1	530.3	
o/w Grants	115.9	161.7	145.2			245.6	160.5	180.9	169.5	154.4	161.8		234.1	490.3	
o/w Concessional loans	21.4	63.9	55.1			54.3	55.9	41.9	39.2	35.0	36.0		40.0	40.0	
Grant-equivalent financing (in percent of GDP) 8/	…	…	…			4.3	2.5	2.4	2.2	1.9	1.8		1.7	1.3	1.6
Grant-equivalent financing (in percent of external financing) 8/	…	…	…			66.8	78.9	81.9	77.0	60.7	47.6		48.0	41.9	45.5
Memorandum items:															
Nominal GDP (Millions of US dollars)	5437.7	5813.4	7002.8			7588.6	8216.4	8968.9	9776.4	10639.8	11670.6		17716.1	40824.8	
Nominal dollar GDP growth	-10.2	6.9	20.5			8.4	8.3	9.2	9.0	8.8	9.7	8.9	8.7	8.7	8.7
PV of PPG external debt (in Millions of US dollars)			1236.7			1360.8	1367.3	1358.9	1361.6	1394.5	1497.8		2197.7	4594.5	
(PVt-PVt-1)/GDPt-1 (in percent)						1.8	0.1	-0.1	0.0	0.3	1.0	0.5	1.0	0.7	1.0
Gross workers' remittances (Millions of US dollars)	1048.2	1085.3	1293.9			1315.7	1404.1	1491.7	1602.8	1725.4	1865.0		2742.2	6319.1	
PV of PPG external debt (in percent of GDP + remittances)	…	…	14.9			15.4	14.2	13.0	12.0	11.3	11.1		10.7	9.7	
PV of PPG external debt (in percent of exports + remittances)	…	…	27.8			29.2	26.2	23.4	21.0	19.3	18.6		17.8	16.1	
Debt service of PPG external debt (in percent of exports + remittances)	…	…	2.0			1.9	2.0	1.9	1.8	2.1	2.2		1.9	2.4	

Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that PV of private sector debt is equivalent to its face value.
5/ Current-year interest payments divided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.
7/ Defined as grants, concessional loans, and debt relief.
8/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

Table 3b.Moldova: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2012-2032
(In percent)

	Projections							
	2012	2013	2014	2015	2016	2017	**2022**	2032
PV of debt-to-GDP+remittances ratio								
Baseline	15	14	13	12	11	11	**11**	10
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2012-2032 1/	15	11	10	9	10	12	**24**	28
A2. New public sector loans on less favorable terms in 2012-2032 2	15	15	14	13	12	12	**14**	17
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	15	15	14	13	12	12	**12**	11
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	15	19	28	26	25	24	**19**	11
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	15	15	14	13	12	12	**11**	10
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	15	19	22	20	19	18	**16**	10
B5. Combination of B1-B4 using one-half standard deviation shocks	15	17	21	19	18	18	**15**	10
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	15	19	17	16	15	15	**14**	13
PV of debt-to-exports+remittances ratio								
Baseline	29	26	23	21	19	19	**18**	16
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2012-2032 1/	29	21	18	17	18	20	**41**	50
A2. New public sector loans on less favorable terms in 2012-2032 2	29	27	24	22	21	21	**24**	27
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	29	26	23	21	19	19	**18**	16
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	29	39	61	56	52	50	**40**	23
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	29	26	23	21	19	19	**18**	16
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	29	35	40	35	33	31	**26**	17
B5. Combination of B1-B4 using one-half standard deviation shocks	29	33	41	37	34	33	**28**	19
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	29	26	23	21	19	19	**18**	16
PV of debt-to-revenue ratio								
Baseline	51	47	42	39	36	36	**36**	36
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2012-2032 1/	51	37	31	29	31	35	**73**	91
A2. New public sector loans on less favorable terms in 2012-2032 2	51	48	44	41	39	40	**49**	61
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	51	49	47	43	40	39	**40**	40
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	51	63	90	84	79	77	**65**	41
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	51	48	45	41	38	38	**38**	38
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	51	61	70	65	61	59	**53**	39
B5. Combination of B1-B4 using one-half standard deviation shocks	51	55	67	62	59	57	**51**	38
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	51	66	60	55	51	50	**51**	51

Table 3b.Moldova: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2012-2032 (continued)
(In percent)

Debt service-to-exports+remittances ratio								
Baseline	2	2	2	2	2	2	**2**	2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2012-2032 1/	2	2	2	1	2	2	**2**	3
A2. New public sector loans on less favorable terms in 2012-2032 2	2	2	2	2	2	2	**1**	3
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	2	2	2	2	2	2	**2**	2
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	2	2	3	3	4	4	**4**	4
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	2	2	2	2	2	2	**2**	2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	2	2	2	2	3	3	**3**	3
B5. Combination of B1-B4 using one-half standard deviation shocks	2	2	2	2	3	3	**3**	3
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	2	2	2	2	2	2	**2**	2
Debt service-to-revenue ratio								
Baseline	3	4	3	3	4	4	**4**	5
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2012-2032 1/	3	3	3	2	3	3	**3**	6
A2. New public sector loans on less favorable terms in 2012-2032 2	3	4	3	3	4	4	**3**	7
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2013-2014	3	4	4	4	4	5	**4**	6
B2. Export value growth at historical average minus one standard deviation in 2013-2014 3/	3	4	4	5	5	6	**7**	7
B3. US dollar GDP deflator at historical average minus one standard deviation in 2013-2014	3	4	4	3	4	4	**4**	6
B4. Net non-debt creating flows at historical average minus one standard deviation in 2013-2014 4/	3	4	4	4	5	5	**6**	6
B5. Combination of B1-B4 using one-half standard deviation shocks	3	3	4	4	5	5	**6**	6
B6. One-time 30 percent nominal depreciation relative to the baseline in 2013 5/	3	5	5	5	5	6	**6**	7
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	4	4	4	4	4	4	**4**	4

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.